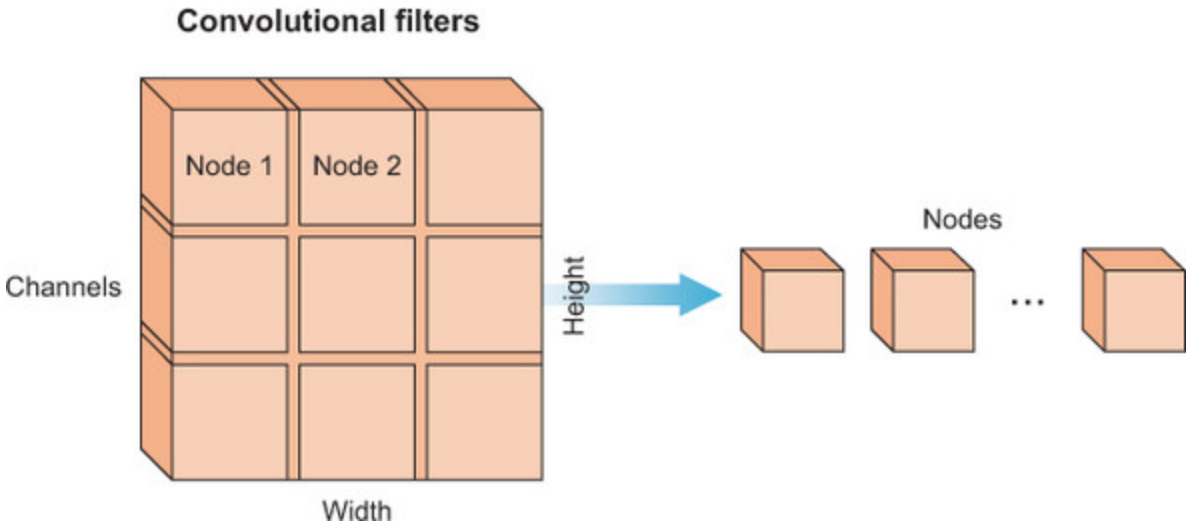


After a raw image has been processed through a few trained CNN layers, we would expect that each position in the feature maps corresponds to particular salient features in the underlying image. For example, we hope the CNNs might learn to detect objects in the image that we can then pass into our relational module to process relations between objects. Each convolutional filter learns a particular feature for each spatial position, so taking all these learned features for a particular (x,y) grid position in an image yields a single vector for that position that encodes all the learned features. We can do this for all the grid positions to collect a set of putative objects in the image, which we can represent as nodes in a graph, except that we do not know the connectivity between the nodes yet. That is what our relational reasoning module will attempt to do.

10.2.3. Self-attention models

There are many possible ways to build a relational module, but as we've discussed, we will implement one based on a self-attention mechanism. We have described the idea at a high level, but it is time we got into the details of implementation. The model we'll build is based on the one described in the paper "Deep reinforcement learning with relational inductive biases" by Vinicius Zambaldi et al. (2019) from DeepMind.

We already discussed the basic framework of a node matrix N:$\mathbb{R}^{n\times f}$ and an edge matrix E:$\mathbb{R}^{n\times n}$, and we discussed the need to process a raw image into